

Data Link Layer & Error Control — Notes

Theory-only revision notes (no numericals) — definitions, MCQ-ready facts, and exam points

1. Data Link Layer

- Converts the physical layer into a **reliable link**.
- Makes the physical layer appear error-free to the Network Layer.
- Moves **frames from one hop to the next**.

2. Responsibilities of Data Link Layer (5)

- **Framing** — bits → frames
- **Physical addressing** — sender/receiver addresses
- **Flow control** — prevents receiver from being overwhelmed
- **Error control** — detects/retransmits damaged or lost frames
- **Access control** — decides which device gets access to a shared link

3. Errors

An error occurs when transmitted data gets **altered, damaged, or lost**.

Causes:

- Noise
- Attenuation
- Delay distortion

4. Types of Errors

- **Single-bit error** — only one bit changes.
- **Burst error** — two or more bits are affected.

5. Error Detection

Determining **whether** an error has occurred. You don't necessarily identify the exact corrupted bit.

6. Error Correction

Determining the **number and location** of corrupted bits and reconstructing the original data.

7. Redundant Bits

Extra bits added along with the actual data to help with **error detection/correction**.

8. ARQ — Automatic Repeat-Request

Data is sent with error-detection information; erroneous data can be **retransmitted**.

9. FEC — Forward Error Correction

The sender sends coded data; the receiver uses the code to determine the most likely correct data **without** sending messages back to the transmitter.

10. Error Detection Schemes

- Parity Check
- CRC
- Checksum

11. Repetition Scheme

The same block of data is transmitted multiple times. The receiver compares the copies to identify an error.

Limitation: not very efficient — certain repeated errors can escape detection.

12. Parity Check

Adds an extra **parity bit** so the number of 1s follows either:

- **Even parity** — even number of 1s
- **Odd parity** — odd number of 1s

Parity is a relatively **weak** error-detection mechanism.

13. Block Sum Check

Extension of the single parity-bit method:

- Each byte gets a **row parity**
- Each bit position gets a **column parity**
- Together these form the **block sum check**

Can detect single-bit errors and increases the chance of detecting burst errors.

14. CRC — Cyclic Redundancy Check ■

- Used especially for **burst-error detection**
- Based on **modulo-2 arithmetic**
- Treats bit strings as polynomial coefficients
- Sender divides data by a predetermined **generator/divisor**
- The remainder becomes the **CRC**
- Receiver divides the received data by the same divisor
- **Zero remainder** → **accepted**
- **Non-zero remainder** → **rejected**

15. CRC Requirements

- CRC has **one bit less** than the divisor.
- Data + CRC should be **exactly divisible** by the divisor.

16. Checksum

- Data is divided into **equal-sized segments**.
 - Segments are added using **1's complement arithmetic**.
 - The result is complemented to produce the checksum.
 - Receiver performs the corresponding check.
 - No error indicated → data accepted; otherwise discarded.
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■ Exam Quick-Recall Summary

Data Link Layer	5 responsibilities: Framing, Physical addressing, Flow control, Error control, Access control
Errors — 2 types	Single-bit, Burst
Error handling	Detection, Correction, ARQ, FEC
Detection methods	Parity, Block Sum Check, CRC, Checksum, Repetition Scheme
CRC	Burst errors • Modulo-2 • Generator/divisor • Remainder • Zero remainder = accepted
Checksum	Equal segments • 1's complement addition • Complement = checksum